C Final Exam Sample Questions

Question 1

Write a C program that asks the user to enter their name, age, and monthly salary, calculates the monthly tax (15% of salary) and net salary (salary minus tax), and displays all information in a well-formatted output.

Question 2

Write a C program that reads two integers from the user, displays the result of all arithmetic operations (`+`, `-`, `*`, `/`, `%`), displays which number is greater using a conditional operator (`?:`), and checks whether the first number is even or odd using the modulus operator.

Question 3

Write a C program that reads a student's marks for 5 subjects, calculates the total and average, assigns a grade (A for 90+, B for 80–89, C for 70–79, D for 60–69, F for below 60), and displays the total, average, grade, and whether the student passed or failed.

Question 4

Write a C program that reads a positive integer `n` from the user and prints a diamond-shaped star pattern using nested loops (increasing then decreasing). Validate that `n` is positive using a `do-while` loop if not, ask again.

Question 5

Write a C program that implements a menu-driven calculator using a `switch` statement with options for addition, subtraction, multiplication, division, and exit. The program must handle division by zero and keep running until the user selects exit.

Question 6

Write a C program that reads 10 integers into an array, finds the largest, smallest, sum, and average, sorts the array in ascending order using bubble sort, and prints the array before and after sorting.

Question 7

Write a C program that reads a sentence from the user, counts the number of vowels, consonants, spaces, and digits, converts the sentence to uppercase without using `toupper()`, counts the total number of words, and prints the sentence reversed.

Question 8

Write a C program with four separate functions: `isPrime(int n)` that returns 1 if n is prime, `factorial(int n)` that returns the factorial using recursion, `fibonacci(int n)` that prints the first n Fibonacci numbers, and `power(int base, int exp)` that returns base raised to exp without using `pow()`. Call all four functions from `main()` using a number entered by the user.

Question 9

Write a C program that defines a structure `Student` with fields name, registration number, age, and GPA. Read details for 3 students, save all records to a file called `students.txt`, read the file back and display all records, and find and display the student with the highest GPA.

Question 10

Write a C program that reads 5 integers into an array, uses a pointer to traverse the array and compute the sum and average, writes a function `findMax(int *arr, int size)` that returns the largest element using pointer arithmetic, writes a function `reverseArray(int *arr, int size)` that reverses the array in place using pointers, and prints the original array, reversed array, sum, average, and maximum.